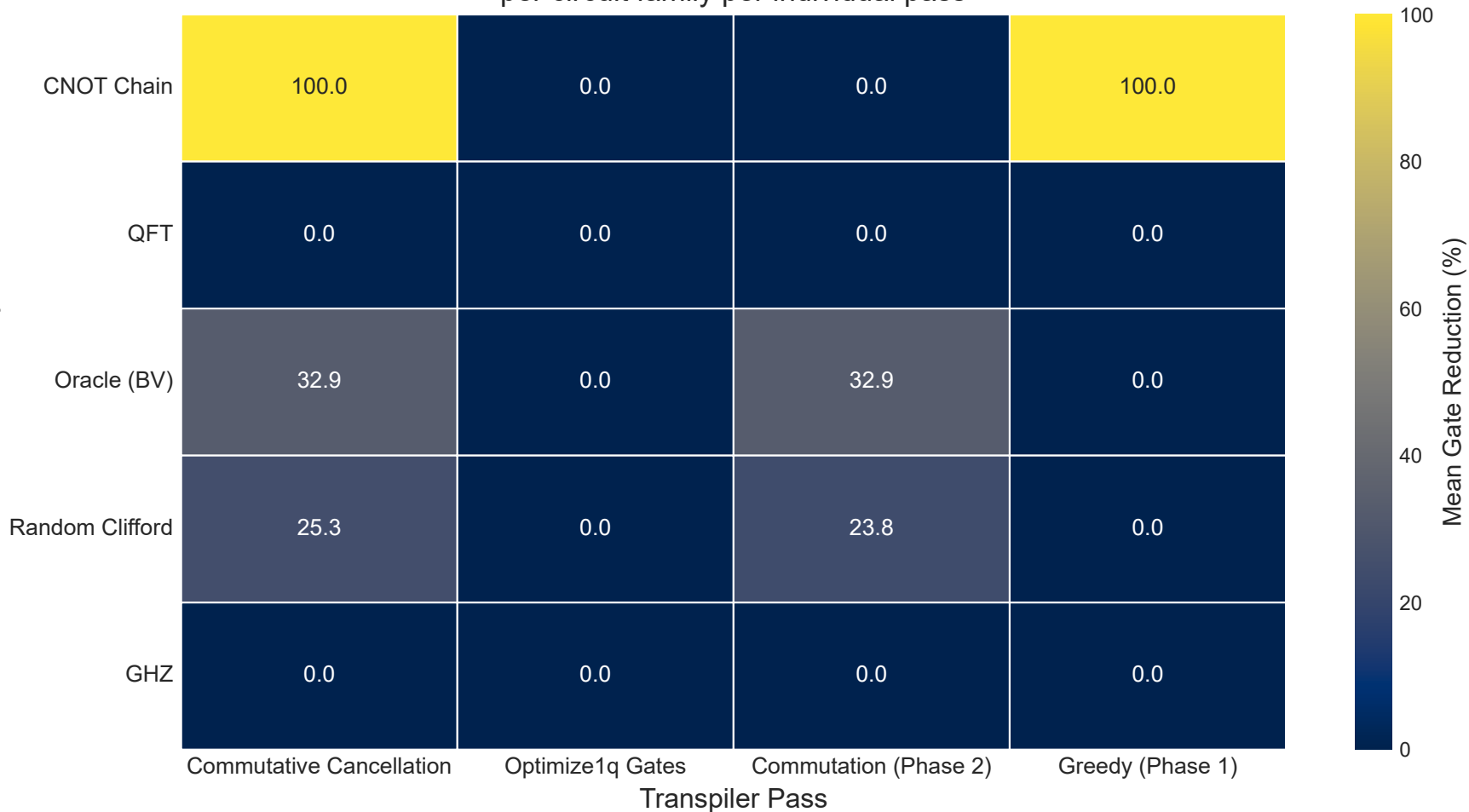


Pass Effectiveness Heatmap — Mean Gate Reduction (%)
per circuit family per individual pass

Circuit Family



Key finding: CommutativeCancellation matches our Phase 2 on Oracle/RandomClifford but both miss QFT/GHZ.
Neither isolated pass explains the full-pipeline advantage (template matching + routing + phase polynomial).